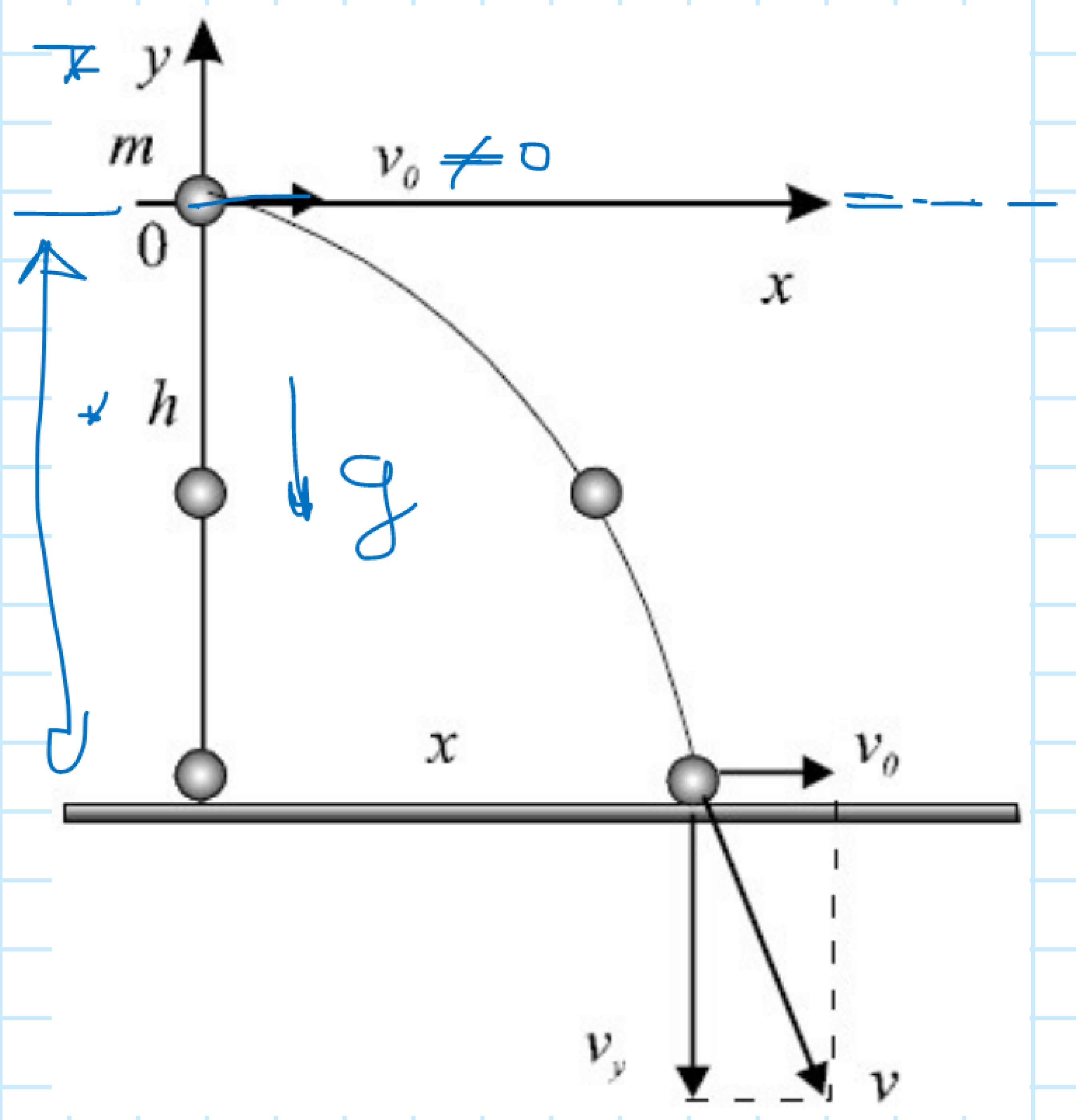
# Ligjerata 3c

## Hedhja horizontale




$\vec{v} = \vec{v}_x + \vec{v}_y$

$v_x = v_0$

$v_y = gt$

$v_x = \frac{dx}{dt} = v_0$

$dx = v_0\, dt \;/\int$

$\int dx = v_0 \int dt$

$x = v_0 t + c$

$t = 0$

$x = 0$

$c = 0$

$\underline{x = v_0 t}$ → komponenta horizontale e rrugës.

$v_y = gt = \frac{dy}{dt}$

$dy = gt\, dt \;/\int$

$\int dy = g \int t\, dt$

$y = g\frac{t^2}{2} + c'$

$t = 0$

$y = 0$

$c' = 0$

$\underline{y = \frac{gt^2}{2}}$ → komponenta vertikale e rrugës

$\vec{v} = \vec{v}_x + \vec{v}_y$

$\vec{v}\cdot\vec{v} = v^2 = (v_x \vec{i} + v_y \vec{j})(v_x \vec{i} + v_y \vec{j})$

$v^2 = v_x^2 \underbrace{\vec{i}\vec{i}}_{1} + v_y^2 \underbrace{\vec{j}\vec{j}}_{1}$

$v^2 = v_x^2 + v_y^2 \;/\sqrt{\ }$

$v = \sqrt{v_x^2 + v_y^2}$

$v_x = v_0 \quad v_y = gt$

$\vec{i}\vec{j} = \vec{j}\vec{k} = \vec{k}\vec{i} = 0$

$\vec{i}\vec{i} = \vec{j}\vec{j} = \vec{k}\vec{k} = 1$

$$\boxed{v = \sqrt{v_0^2 + g^2 t^2}}$$